

Lesson 6 Proving Universal and Existence Statements

We have introduced math proofs and learned the basic elements of logic and sets. We will study increasingly complex proofs, with attention to *logic*, *definitions*, and *the use of axioms and proven theorems*. In this lesson, we focus on proving (and disproving) a variety of universal and existence statements.

Proposition 1 The sum of two even integers is even.

Proof Let x and y be two (arbitrary) even integers.

Then $x = 2n$ and $y = 2m$ for some integers n and m , and

$$x + y = 2n + 2m = 2(n + m)$$

Since $n + m$ is an integer, $x + y$ is even.

We highlight a few important points.

- The proposition is a *universal conditional*: “For all $x, y \in \mathbb{Z}$, if x and y are even, then $x + y$ is even.” In our proof, we use arbitrary x and y . A universal statement CANNOT be proved by examples! It would be a *big mistake* to prove this proposition with an example $2 + 4 = 6$.
- The proposition is a conditional $p \rightarrow q$, and our proof is a *direct* proof: we assume that p is true and show after a chain of reasoning, q is true.
- We start each sentence with a new line. As beginners, we may do this for clarity. When we are comfortable, we should write a proof in a paragraph.

Proposition 2 Let a, b, c, d be integers. If $a|b$ and $c|d$, then $ac|bd$.

Proof If $a|b$ and $c|d$, then $b = an$ and $d = cm$ for some $n, m \in \mathbb{Z}$.

Since $bd = (an)(cm) = (ac)(nm)$ and nm is an integer, ac divides bd .

So far, our proofs are based mainly on definitions. Next, we will prove a theorem using an axiom, the Division Algorithm (Lesson 5).

Theorem 3 An integer is even or odd but not both.

Proof Let x be an integer. By the Division Algorithm, there are *unique* integers q and r such that $x = 2q + r$, $0 \leq r < 2$. So, $x = 2q$ or $x = 2q + 1$ for some integer q but not both, and x is even or odd, but not both.

To prove a universal statement, we may have to consider *different cases*.

Proposition 4 If x and y are integers with the same parity, then $x + y$ is even.

Proof We consider two cases. If x and y are even, we are done by Proposition 1. If x and y are odd, then $x = 2n + 1$ and $y = 2m + 1$ for some $n, m \in \mathbb{Z}$,

$$x + y = 2n + 1 + 2m + 1 = 2(n + m + 1)$$

Since $n + m + 1$ is an integer, $x + y$ is even.

Proposition 5 If x and y are integers with different parity, then $x + y$ is odd.

Proof Without loss of generality, let x be even and y be odd, then $x = 2n$ and $y = 2m + 1$ for some integers n, m , and $x + y = 2n + 2m + 1 = 2(n + m) + 1$. Since $n + m$ is an integer, $x + y$ is odd.

WLOG is applicable because the two cases, (i) x even and y odd, and (ii) x odd and y even, are *essentially* the same: one even, one odd, and may be merged. However, WLOG cannot be used in the proof of Proposition 4; the two cases, both even and both odd, are *not* essentially the same and cannot be merged.

So far, we have proved universal statements; the keyword is *arbitrary* or *general*. We move on to *existence* statements: “ $\exists x \in A, s(x)$ ”, we need only to give an *example* of x such that $s(x)$ is true, no more!

Proposition 6 There exists a natural number x such that $2^x < x^2$.

Proof We prove with the natural number $x = 3$: $2^3 < 3^2$.

Let’s look at a statement with more than one quantifier.

Proposition 7 For every integer x , there is an integer y such that $x + y = 0$

Proof Let x be an integer, then $y = -x$ is an integer and $x + y = x + (-x) = 0$.

In this proof, x is arbitrary while $y = -x$ is specific given the arbitrary x . There may be another y other than $-x$ such that $x + y = 0$, we don’t care, unless we have a stronger statement, “there is a *unique* integer y such that $x + y = 0$ ”. Then our proof continues to show uniqueness:

If integers y and z satisfy $x + y = 0$ and $x + z = 0$, then $x + y = x + z$, and $y = z$ (subtract x from both sides). So there is a unique y such that $x + y = 0$.

We move on to **disproving** a statement, equivalent to proving its negation.

- To disprove a universal “ $\forall x \in A, s(x)$ ”, we prove its negation, the existence statement “ $\exists x \in A, \sim s(x)$ ”, with a **counterexample** of $s(x)$, that is, x such that $\sim s(x)$ is true or $s(x)$ is false.
- To disprove an existence statement “ $\exists x \in A, s(x)$ ”, we prove its negation, the universal statement “ $\forall x \in A, \sim s(x)$ ”.

We disprove “For all real numbers x , $x^2 \geq x$ ” with a counterexample: $x = 0.5$ is a real number, but $x^2 = 0.25 < 0.5 = x$.

To disprove “There is a real number x such that $x^2 + 2x + 2 = 0$ ”, we prove its negation, the universal statement,

For every real number x , $x^2 + 2x + 2 \neq 0$

Proof For all real numbers x ,

$$x^2 + 2x + 2 = (x + 1)^2 + 1 \geq 1 > 0$$

This example shows that to carry out a mathematical proof, you need “math knowledge”, in this case, completing the square, *in addition to* knowing what it means to give a proof.